IN A 2-CARTESIAN SQUARE WITH FULLY FAITHFUL LOCALLY ESSENTIALLY SURJECTIVE LEG, STACK-IN-SETOIDS DESCENDS

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Explicit 2-fibre product model. categories-lemma-2-product-categories-over-C (cross-chapter, named). Identifies the fibre product T_2 = S_2 x_{S_1} T_1 and the morphism sets used to read off Aut(y').

Source reference. Original reference 0040: focused-seeds/geometry/S9-stk-stack-in-setoids-descent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let*

$$\begin{array}{ccc} \mathcal{T}_2 & \longrightarrow & \mathcal{T}_1 \\ \downarrow & & \downarrow{\scriptstyle G} \\ \mathcal{S}_2 & \xrightarrow{F} & \mathcal{S}_1 \end{array}$$

be a 2-cartesian diagram of stacks in groupoids over $\mathcal{C}$. If

(1) *$F : \mathcal{S}_2 \to \mathcal{S}_1$ is fully faithful,*
(2) *for every $U \in \mathrm{Ob}(\mathcal{C})$ and $x \in \mathrm{Ob}((\mathcal{S}_1)_U)$ there exists a covering $\{U_i \to U\}$ such that $x|_{U_i}$ is in the essential image of $F : (\mathcal{S}_2)_{U_i} \to (\mathcal{S}_1)_{U_i}$, and*
(3) *$\mathcal{T}_2$ is a stack in setoids.*

then $\mathcal{T}_1$ is a stack in setoids.

Proof. We may assume that $\mathcal{T}_2$ is the category $\mathcal{S}_2 \times_{\mathcal{S}_1} \mathcal{T}_1$ described in Categories, Lemma 0040. Pick $U \in \mathrm{Ob}(\mathcal{C})$ and $y \in \mathrm{Ob}((\mathcal{T}_1)_U)$. We have to show that the sheaf $Aut(y)$ on $\mathcal{C}/U$ is trivial. To to this we may replace U by the members of a covering of U. Hence by assumption (2) we may assume that there exists an object $x \in \mathrm{Ob}((\mathcal{S}_2)_U)$ and an isomorphism $f : F(x) \to G(y)$. Then $y' = (U, x, y, f)$ is an object of $\mathcal{T}_2$ over U which is mapped to y under the projection $\mathcal{T}_2 \to \mathcal{T}_1$. Because F is fully faithful by (1) the map $Aut(y') \to Aut(y)$ is surjective, use the explicit description of morphisms in $\mathcal{T}_2$ in Categories, Lemma 0040. Since by (3) the sheaf $Aut(y')$ is trivial we get the result of the lemma. □

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.